

AGRI-ELECTRONICS

The Emerging Frontier of Advanced Research

Farmers toil hard in the field to feed the world. Thankfully, a few R&D labs in India are determined to make farming easier by automating the various processes through use of electronics and IT. Take a look at some of these interesting innovations



■ DR NABARUN BHATTACHARYYA

Agri-electronics is an emerging and multi-disciplinary frontier of advanced research that takes convergent and holistic look at all the related areas in the agricultural sector. It ensures optimal induction of IT and electronics for improvement in crop productivity, quality and value.

R&D in agri-electronics is a potential vehicle for digitisation of green

revolution. It promises a catalytic role in national agricultural development in terms of diversification, enhancing productivity, adding value, capturing markets, mixing farm and off-farm income, entering and creating marketing chains, improving food quality and safety, and balancing ecological interests.

Scope and challenges of agri-electronics

The latest practice adopted worldwide is precision agriculture. It implies knowledge-based agriculture employing the

latest techniques of science and technology bringing together diverse fields—agronomy, plant science, soil science, entomology, metrology, weed science, plant pathology, ecology, water technology, environment and economics—to deal with complex issues of production, distribution, pricing, marketing, etc.

Fig. 1 shows the scope of agri-electronics. To deal with cross-disciplinary complexities, multitudes of technologies like bio-electronics, sensors, software, embedded systems and soft computing need to be harnessed to develop efficient agri-electronics solutions. Management of new technologies for agri-electronics research is one of the stiffest challenges. Agri-electronics research must be carefully planned and modulated to take account of present and future trends in agriculture, biotechnology, and information and communication technology (ICT).

Partnerships and networking are essential for a multi-disciplinary field like agri-electronics. It is imperative that agri-electronics research be conducted through alliances. Stakeholders and researchers from different organisations need to collaborate to identify the needs for specific kinds of knowledge, to implement research projects, and to disseminate results.

Bringing different interest groups together (such as policy-makers, scientists, traders, farmers, producers, governments, industries and consumers) is the